

Lab 11 Counting

Objective

Solving exercises from the textbook.

Current Lab Learning Outcomes (LLO)

By completion of the lab, the students should be able to

1. Understand basics of counting.
2. Solve shorter/easier or longer / harder problems given in the textbook.

Lab Requirements

Students allowed using their lecture notes in the lab and use blackboard slides in order to solve the exercises.

Lab Assessment

- 1- Divide students to groups and let them to solve the given example.
- 2- Discuss the answers with the groups and write on board the optimal solution.

Lab Description

In this lab, students learn how to apply the basic counting principles, the pigeonhole principle, and permutations and combinations to solve counting problems

1. There are 18 mathematics majors and 325 computer science majors at a college.
 - a) In how many ways can two representatives be picked so that one is mathematics major and the other is a computer science major?
 - b) In how many ways can one representative be picked who is either mathematics major or a computer science major?
2. Six different airlines fly from New York to Denver and seven fly from Denver to San Francisco. How many different pairs of airlines can you choose on which to book a trip from New York to San Francisco via Denver, when you pick an airline for the flight to Denver and an airline for the continuation flight to San Francisco?
3. In how many ways can a photographer at a wedding arrange six people in a row, including the bride and groom, if
 - a) The bride must be next to the groom.
 - b) The bride is not next to the groom.
 - c) The bride is positioned somewhere to the left of the groom.
4. How many permutations of $\{a, b, c, d, e, f, g\}$ end with a ?

5. How many possibilities are there for the win, place, and show (first, second, and third) positions in a horse race with 12 horses if all orders of finish are possible?
6. A club has 25 members.
 - a) How many ways are there to choose four members of the club to serve on an executive committee?
 - b) How many ways are there to choose a president, vice president, secretary, and treasurer of the club, where no person can hold more than one office?
7. Suppose that a department contains 10 men and 15 women. How many ways are there to form a committee with six members if it must have the same number of men and women?
8. A multiple-choice test contains 10 questions. There are four possible answers for each question.
 - a) In how many ways can a student answer the questions on the test if the student answers every question?
 - b) In how many ways can a student answer the questions on the test if the student can leave answers blank?
10. How many different three-letter initials are there that begin with an A?
11. How many bit strings of length ten both begin and end with a 1?
12. How many strings of three decimal digits
 - a) Do not contain the same digit three times?
 - b) Begin with an odd digit?
 - c) Have exactly two digits that are 4's?
13. How many strings of eight English letters are there
 - a) That contain no vowels, if letters can be repeated.
 - b) That contain no vowels, if letters cannot be repeated.
 - c) That start with a vowel, if letters can be repeated.
14. How many bit strings of length ten both begin and end with a 1?
15. How many bit strings of length 10 either begin with three 0's or end with two 0's?
16. A drawer contains a dozen brown socks and a dozen black socks, all unmatched. A man takes socks out at random in the dark.
 - a) How many socks must he take out to be sure that he has at least two socks of the same color?
 - b) How many socks must he take out to be sure that he has at least two black socks?
17. There are 38 different time periods during which classes at a university can be scheduled. If there are 677 different classes, how many different rooms will be needed?
18. Suppose that every student in a discrete mathematics class of 25 students is a freshman, a sophomore, or a junior.
 - a) Show that there are at least nine freshmen, at least nine sophomores, or at least nine juniors in the class.
 - b) Show that there are at least three freshmen, at least 19 sophomores, or at least five juniors in the class.

